

Intro to Koopman Operators

Consider a discrete-time dynamical system:

$$x_{k+1} = \underbrace{F(x_k)}_{F: \mathbb{R}^d \rightarrow \mathbb{R}^d}, \quad k=0,1,2,\dots$$

The classical approach to nonlinear dynamics focuses on state-space trajectories and the geometry/topology of the state-space.

Koopman Operator theory offers an alt. approach based on observables/measurements. A measurement of the system's state is modelled by an **observable** function

$$g: \mathbb{R}^d \rightarrow \mathbb{R}.$$

The **Koopman operator** acts on observables:

$$[K_g](x) = g(F(x)) = [g \circ F](x).$$

A sequence of measurements along a single trajectory starting at x_0 is given by

$$g(x_n) = [g \circ \underbrace{F \circ \dots \circ F}_{n \text{ times}}](x_0) = [K^n g](x_0).$$

So powers of the Koopman operator describe the dynamics of the measurements.

The crucial property of the Koopman operator is that it is linear w.r.t. observables:

$$\begin{aligned} [K(g + \alpha f)](x) &= g(F(x)) + \alpha f(F(x)) \\ &= [Kg](x) + \alpha [Kf](x). \end{aligned}$$

Key Idea: Study linear dynamics on a space of observables, defined by

$$g_{k+1} = K g_k, \quad k=0, 1, 2, \dots$$

Q: What space of observables should we use?

Q: What can we learn about the original dynamical system from the Koopman op.?

We will unpack these two questions over the next several lectures w/a series of examples and theoretical results.

To address the first question, first note that the space of observables must typically be infinite-dimensional.

Example. Cubic map

Take $F(x) = x - x^3$ and $g(x) = x$.

$$\Rightarrow [K_g](x) = x - x^3, [K_g^2](x) = x - x^3 - (x - x^3)^3, \dots$$

The dimension of $\text{span}\{g, K_g, K_g^2, \dots, K_g^n\}$ grows rapidly as n increases, since each application of K leads to a higher power of x , which is lin. indep. from prev. terms.

To address the second question, we'll make use of spectral decompositions of the Koopman operator.

Koopman Mode Decomposition (Version 1)

Suppose that K has some ^(lin. indep.) eigenpairs:

$$K\phi_j = \lambda_j \phi_j, \quad j = 1, \dots, n.$$

Then, any $g \in \text{span}\{\phi_1, \dots, \phi_n\}$ has some coordinates $c_1, \dots, c_n$ such that

$$g(x) = c_1 \phi_1(x) + \dots + c_n \phi_n(x),$$

$$[K^L g](x) = c_1 \lambda_1^L \phi_1(x) + \dots + c_n \lambda_n^L \phi_n(x).$$

This Koopman Mode Decomposition (KMD) separates the observable dynamics into three fundamental components:

$c_1, \dots, c_n$ - coordinates of g w.r.t. $\{\phi_1, \dots, \phi_n\}$
describe influence of observable.

$\lambda_1^d, \dots, \lambda_n^d$ - eigenvalues of K describe how
expansion modes evolve in time.

$\phi_1, \dots, \phi_n$ - eigenfunctions of K encode
features of state-space dynamics.

Since the Koopman Operator usually
must act on an infinite-dimensional
space of observables, it is possible that
no finite-dimensional invariant subspace
exists. We must be careful in how to proceed.

Example. Bilateral Shift

Consider the operator $K: \ell^2 \rightarrow \ell^2$, where

$$K e_n = e_{n-1}, \quad e_n^{(j)} = \delta_{nj}$$

and $\ell^2 = \left\{ \{v^{(j)}\}_{j=-\infty}^{+\infty} \mid \sum_{j=-\infty}^{+\infty} |v^{(j)}|^2 < \infty \right\}$. You may think of ℓ^2 as an inf-dim Euclidean space, and K as the $\infty \times \infty$ unitary matrix

$$Kv = \begin{bmatrix} \vdots & & & & \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & 0 & 1 \\ & & & & \ddots & \ddots \\ & & & & & 0 & 1 \\ & & & & & & \ddots & \ddots \\ & & & & & & & 0 & 1 \\ & & & & & & & & \ddots & \ddots \end{bmatrix} \begin{bmatrix} \vdots \\ v^{(-n)} \\ \vdots \\ v^{(n)} \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ v^{(-n+1)} \\ \vdots \\ v^{(n+1)} \\ \vdots \end{bmatrix}.$$

The solutions to $Kv = \lambda v$ have form

$$v^{(j)} = \lambda^j, \quad j = -\infty, \dots, +\infty$$

with eigenvalue $\lambda \in \mathbb{C} \setminus \{0\}$. However,

$$\sum_{j=-\infty}^{+\infty} |v^{(j)}|^2 = \infty,$$

for any $\lambda \in \mathbb{C} \setminus \{0\}$, so there are no solutions in the inf-dim space ℓ^2 .

$\Rightarrow K: \ell^2 \rightarrow \ell^2$ is unitary but has no finite-dimensional invariant subspace.